Quiz 1

Name: **Roll Number**.................

- Answer precisely in the space given. No overwriting. Make assumptions, if really ambiguous.
- More than one option may be correct in many questions. Notations follow the lecture.
- Questions 1-10: 3 Points; Questions 11-12: 7.5 Points. Total 45 Points. ∘

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1. Consider the vector $\mathbf{x} \in R^4$ as $[-4, 3, 0, 0]^T$. Find the following norms:
   (i) L_0 ————
   (ii) L_1 ————
   (iii) L_2 ————
   (iv) L_∞ ————

2. Solution to the optimization problem: "Minimize $\mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}$ such that $||\mathbf{w}||_2^2 = 1$" is

   (A) ——————————————- Eigen Vector of (B) ——————————————

   Fill (A) and (B) with appropriate options from below.

   (A) (i) Largest (ii) Second Largest (iii) Smallest (iv) Second Smallest

   (B) (i) $\mathbf{X}^T\mathbf{X}$ (ii) $\mathbf{X}\mathbf{X}^T$ (iii) $\mathbf{X}$ (iv) $\mathbf{X}^T$

3. Consider a dataset of 1000 frontal aligned face images each of size 100×100. We are interested in finding Eigen faces. Which of the following are true?
   - Covariance matrix is of size 100×100
   - Covariance matrix is of size 10000×10000
   - Covariance Matrix is of size 1000×1000
   - Practical rank of the covariance matrix will be strictly less than 100.
   - Practical rank of the covariance matrix will be far lesser than $min(N, d)$

4. Consider a problem of deciding an email as SPAM or NON-SPAM. We have a trained classifier/solution that scores each email as $\mathbf{w}^T\mathbf{x}$. However, our job will be to define a threshold so that we can say $\mathbf{w}^T\mathbf{x} \geq \theta$ for SPAMs

   For the 10 emails that we have the scores (i.e., $\mathbf{w}^T\mathbf{x}$) are

   $$0.7, 0.65, 0.92, 0.75, 0.45, 0.8, 0.3, 0.8, 0.1, 0.1$$

   and we know the truth/label as (SPAM (+) and NON-SPAM (-))

   $$+, -, +, +, -, +, +, +, -, -$$

   For $\theta = 0.5$, find the "Precision" and "Recall".

   Ans: Precision: —————-; Recall: ——————-

5. Consider a dataset ($\mathbf{x}$, y) with Training and Test as:

   $$\mathcal{D}_{Train} = \{([+2.0, 0.5]^T, +), ([-3.0, +0.5]^T, -), ([0.0, 3.0]^T, +), ([0.0, -3.0]^T, -)\}$$
   $$\mathcal{D}_{Test} = \{([0, 0]^T, +), ([+1, +1]^T, +), ([-1, +1]^T, +), ([-1, -1]^T, -), ([+1, -1]^T, -)\}$$

   What is the accuracy of a 1-NN classifier with distance function as Manhattan distance? (Manhattan distance between $\mathbf{p}$ and $\mathbf{q}$ is the L1 norm of $\mathbf{p} - \mathbf{q}$

6. We compute SVD of a $m \times n$ with $m > n$ matrix as $A \rightarrow UDV^T$. Fill with the right words "(a) ——————— of U are orthogonal; (b)——————— of V are orthogonal and D_{ii} are eigen values of (c) ———————"

   (a) Columns, Rows
   (b) Columns, Rows
   (c) AA^T; A^TA; Both AA^T and A^TA

7. We are given a set of samples:

$$\mathcal{D} = \{[2.0, 3.0]^T, [3.0, 2.0]^T, [2.0, -3.0]^T, [0.0, 3.0]^T, [0, 0]^T, [+1, +1]^T[-1, +1]^T\}$$

   Normalize and create a new data set $\mathcal{D}'$ such that mean of this set is $[0, 0]^T$

   $\mathcal{D}' =$

8. Consider the linear regression with N samples $(\mathbf{x}_i, y_i)$ and prediction as $\hat{y} = \mathbf{w}^T\mathbf{x}$.

   Write the MSE Loss function that we need to minimize: ———————

9. You know the objective function that we need to optimize/maximize for PCA as

$$\max_{\mathbf{v}} \ \mathbf{v}^T \Sigma \mathbf{v}$$

   Since this is a maximization problem, we can use gradient ascent (not descent) to solve this. Suggest an update equation to find the solution. i.e., find $\mathbf{v}^{k+1}$ from $\mathbf{v}^k$

   (a) $\mathbf{v}_{k+1} = \eta \Sigma \mathbf{v}_k$
   (b) $\mathbf{v}_{k+1} = (I + \eta\Sigma)\mathbf{v}_k$
   (c) $\mathbf{v}_{k+1} = (I - \eta\Sigma)\mathbf{v}_k$
   (d) None of these

   (Note: the constraint $||\mathbf{v}|| = 1$ can be enforced as a seperate step in each iteration.)

10. Given a set of 2D points X on a line that makes 45 degree to the x-axis:

$$X = \{[-2, 2]^T, [-3, 3]^,[-4, 4]^T, [-5, 5]^T[-6, 6]^T\}$$

   We compute the covariance matrix, and its eigen values and eigen vectors. Then identify the correct answer.

   (a) $\lambda_2 = 0$
   (b) $\lambda_1 = \lambda_2$
   (c) $\lambda_1 = -1$
   (d) Σ is singular
   (e) none of the above

11. Consider a loss/objective function that we want to minimize for obtaining w^* as

$$w^* = \arg\min_w \ \mathcal{L}(w)$$

$$\text{where } \mathcal{L}(w) = 2w^2 - 4w + 7$$

We want to find the minima of the function using gradient descent method. Write update equation for computing w^{k+1} from w^k, where k is the iteration number.

Ans (i): ____________

If $w^0 = 2.0$, what should be w^1 and w^2 if the learning rate $\eta = 0.2$?

Ans (ii): w^1 = ____________ and w^2 = ____________

At the optima, what will be the loss value, i.e., $\mathcal{L}(w*)$

Ans (iii): $\mathcal{L}(w^*)$ = ____________ [1.5 + 3.0 + 3.0 = 2.5]

12. We know the closed form solution to the regression as

$$\mathbf{w}^* = (\mathbf{X}^{\mathbf{T}}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

In a practical problem, Raju collected data $\mathcal{D} = \{(x, y)\}$ as {(1,2), (2,3), (3,4)}. (note that x is a simple scalar; and on a quick look, you know the solution as $y = x + 1$. Let us verify what we learned in the class really give the same answer.)

By constructing $\mathbf{X}$ and $\mathbf{y}$, compute $\mathbf{w}^*$

Rough Work (will not be graded)

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- Questions 1-10: 3 Points; Questions 11-12: 7.5 Points. Total 45 Points. ‡

1. Consider the vector $\mathbf{x} \in R^4$ as $[-6, 3, 2, 0]^T$. Find the following norms:
(i) L_0 ————
(ii) L_1 ————
(iii) L_2 ————
(iv) L_∞ ————

2. Solution to the optimization problem: "Maximize $\mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}$ such that $||\mathbf{w}||_2^2 = 1$" is

   (A) ——————————————- Eigen Vector of (B) ——————————————

   Fill (A) and (B) with appropriate options from below.

   (A) (i) Largest (ii) Second Largest (iii) Smallest (iv) Second Smallest

   (B) (i) $\mathbf{X}^T\mathbf{X}$ (ii) $\mathbf{X}\mathbf{X}^T$ (iii) $\mathbf{X}$ (iv) $\mathbf{X}^T$

3. Consider a dataset of 1000 frontal aligned face images each of size 100×100. We are interested in finding Eigen faces. Raju who implemented the algorithm by mistake duplicated his own face photo 1000 times to create the dataset.

   Which of the following are true?

   (a) Even if all images were identical, mean will be a "blurred" image due to the averaging effect.
   (b) If all images were identical, mean will be the same as that of Raju's original face.
   (c) All elements of the covariance matrix will be zero.
   (d) Covariance matrix is of size 100×100
   (e) If Raju had modified each of the 1000 (not just duplicating), by changing contrast, etc., covariance matrix will have strictly only one non-zero eigen value.

4. Consider a problem of deciding an email as SPAM or NON-SPAM. We have a trained classifier/solution that scores each email as $\mathbf{w}^T\mathbf{x}$. However, our job will be to define a threshold so that we can say $\mathbf{w}^T\mathbf{x} \geq \theta$ for SPAMs

   For the 10 emails that we have the scores (i.e., $\mathbf{w}^T\mathbf{x}$) are

   $$0.7, 0.65, 0.92, 0.75, 0.45, 0.8, 0.3, 0.8, 0.1, 0.1$$

   and we know the truth/label as (SPAM (+) and NON-SPAM (-))

   $$+, -, +, +, -, +, +, +, -, -$$

   For $\theta = 0.2$, find the "Precision" and "Recall".

   Ans: Precision: —————-; Recall: ——————-

5. Consider a dataset $(\mathbf{x}, \mathrm{y})$ with Training and Test as:

   $$\mathcal{D}_{Train} = \{([+2.0, 0.5]^T, +), ([-3.0, +0.5]^T, -), ([0.0, 3.0]^T, +), ([0.0, -3.0]^T, -)\}$$
   $$\mathcal{D}_{Test} = \{([0, 0]^T, +), ([+1, +1]^T, +), ([-1, +1]^T, +), ([-1, -1]^T, -), ([+1, -1]^T, -)\}$$

   What is the accuracy of a 3-NN classifier with distance function as Manhattan distance? (Manhattan distance between $\mathbf{p}$ and $\mathbf{q}$ is the L1 norm of $\mathbf{p} - \mathbf{q}$

6. We compute SVD of a $m \times n$ with $m > n$ matrix as $A \rightarrow UDV^T$. Fill with the right words "(a) ______________ of U are orthogonal; (b)______________ of V are orthogonal and D_{ii} are eigen values of (c) ________________"

   (a) Columns, Rows
   (b) Columns, Rows
   (c) AA^T; A^TA; Both AA^T and A^TA

7. We are given a set of samples:

$$\mathcal{D} = \{[2.0, 3.0]^T, [3.0, 2.0]^T, [2.0, -3.0]^T, [0.0, 3.0]^T, [0, 0]^T, [+1, +1]^T[-1, +1]^T\}$$

   Normalize and create a new data set $\mathcal{D}'$ such that mean of this set is $[0, 0]^T$

   $\mathcal{D}' =$

8. Consider the Ridge linear regression with N samples $(\mathbf{x}_i, y_i)$ and prediction as $\hat{y} = \mathbf{w}^T\mathbf{x}$.

   Write the Loss function that we need to minimize: ______________

9. You know the objective function that we need to optimize/maximize for PCA as

$$\max_{\mathbf{v}} \ \mathbf{v}^T\Sigma\mathbf{v}$$

   Since this is a maximization problem, we can use gradient ascent (not descent) to solve this. Suggest an update equation to find the solution. i.e., find $\mathbf{v}^{k+1}$ from $\mathbf{v}^k$

   (a) $\mathbf{v}_{k+1} = \eta\Sigma\mathbf{v}_k$
   (b) $\mathbf{v}_{k+1} = (I + \eta\Sigma)\mathbf{v}_k$
   (c) $\mathbf{v}_{k+1} = (I - \eta\Sigma)\mathbf{v}_k$
   (d) None of these

   (Note: the constraint $||\mathbf{v}|| = 1$ can be enforced as a separate step in each iteration.)

10. Given a set of 2D points X on a line that makes 45 degree to the x-axis:

$$X = \{[-2, 2]^T, [-3, 3]'[-4, 4]^T, [-5, 5]^T[-6, 6]^T\}$$

   We compute the covariance matrix, and its eigen values and eigen vectors. Then identify the correct answer.

   (a) $\lambda_2 = 0$
   (b) $\lambda_1 = \lambda_2$
   (c) $\lambda_1 = -1$
   (d) Σ is singular
   (e) none of the above

11. Consider a loss/objective function that we want to minimize for obtaining w^* as

$$w^* = \arg\min_{w} \ \mathcal{L}(w)$$

$$\text{where } \mathcal{L}(w) = 2w^2 - 4w + 7$$

We want to find the minima of the function using gradient descent method. Write update equation for computing w^{k+1} from w^k, where k is the iteration number.

Ans (i): ____________

If $w^0 = 0.0$, what should be w^1 and w^2 if the learning rate $\eta = 0.2$?

Ans (ii): $w^1 =$ ____________ and $w^2 =$ ____________

At the optima, what will be the loss value, i.e., $\mathcal{L}(w*)$

Ans (iii): $\mathcal{L}(w^*) =$ ____________ [1.5 + 3.0 + 3.0 = 7.5]

12. We know the closed form solution to the regression as

$$\mathbf{w}^* = (\mathbf{X}^{\mathbf{T}}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

In a practical problem, Raju collected data $\mathcal{D} = \{(x, y)\}$ as {(1,-1), (2,0), (3,1)}. (note that x is a simple scalar; and on a quick look, you know the solution as $y = x - 2$. Let us verify what we learned in the class really give the same answer.)

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1. Consider the vector $\mathbf{x} \in R^4$ as $[-4, 2, 0, 4]^T$. Find the following norms:
(i) L_0 ————
(ii) L_1 ————
(iii) L_2 ————
(iv) L_∞ ————

2. Solution to the optimization problem: "Minimize $\mathbf{w}^T\mathbf{X}\mathbf{X}^T\mathbf{w}$ such that $||\mathbf{w}||_2^2 = 1$" is

   (A) ——————————————- Eigen Vector of (B) ——————————————

   Fill (A) and (B) with appropriate options from below.

   (A) (i) Largest (ii) Second Largest (iii) Smallest (iv) Second Smallest

   (B) (i) $\mathbf{X}^T\mathbf{X}$ (ii) $\mathbf{X}\mathbf{X}^T$ (iii) $\mathbf{X}$ (iv) $\mathbf{X}^T$

3. Consider a dataset of 1000 frontal aligned face images each of size 100×100. We are interested in finding Eigen faces. Which of the following are true?

   (a) Only the initial couple of hundred eigen values are significant, beyond that they can be neglected, since they are very small, even if non-zero.
   (b) Only the initial couple of thousand eigen values are significant, beyond that they can be neglected, since they are very small, even if non-zero.
   (c) If the 1000 images were from 5 students, (say 200 images of each), practical rank of the covariance matrix will be closed to 5 and far from 200.
   (d) If the 1000 images were from 5 students, (say 200 images of each), practical rank of the covariance matrix will be closed to 200 and far from 5.
   (e) If all images were rotated by 180° (i.e., faces are now upside down) or any fixed arbitrary angle, rank of the covariance matrix won't change
   (f) If all images were rotated independently by arbitrary but different angles, rank of the covariance matrix won't change

4. Consider a problem of deciding an email as SPAM or NON-SPAM. We have a trained classifier/solution that scores each email as $\mathbf{w}^T\mathbf{x}$. However, our job will be to define a threshold so that we can say $\mathbf{w}^T\mathbf{x} \geq \theta$ for SPAMs

   For the 10 emails that we have the scores (i.e., $\mathbf{w}^T\mathbf{x}$) are

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   and we know the truth/label as (SPAM (+) and NON-SPAM (-))

   $$+, -, +, +, -, +, +, +, -, -$$

   For $\theta = 0.6$, find the "Precision" and "Recall".

   Ans: Precision: ————-; Recall: ————-

   Consider a dataset ($\mathbf{x}$, y) with Training and Test as:

$$\mathcal{D}_{Train} = \{([+2.0, 0.5]^T, +), ([-3.0, +0.5]^T, -), ([0.0, 3.0]^T, +), ([0.0, -3.0]^T, -)\}$$
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What is the accuracy of a 1-NN classifier with distance function as Euclidean distance? (Euclidean distance between $\mathbf{p}$ and $\mathbf{q}$ is the L2 norm of $\mathbf{p} - \mathbf{q}$

5. We compute SVD of a $m \times n$ with $m < n$ matrix as $A \rightarrow UDV^T$. Fill with the right words "(a) ________________ of U are orthogonal; (b)________________ of V are orthogonal and D_{ii} are eigen values of (c) ________________"
   (a) Columns, Rows
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6. We are given a set of samples:
$$\mathcal{D} = \{[2.0, 3.0]^T, [3.0, 2.0]^T, [2.0, -3.0]^T, [0.0, 3.0]^T, [0, 0]^T, [+1, +1]^T[-1, +1]^T\}$$
   Normalize and create a new data set $\mathcal{D}'$ such that mean of this set is $[0, 0]^T$

   $\mathcal{D}' =$

7. Consider the LASSO linear regression with N samples $(\mathbf{x}_i, y_i)$ and prediction as $\hat{y} = \mathbf{w}^T\mathbf{x}$.

   Write the Loss function that we need to minimize: ________________

8. You know the objective function that we need to optimize/maximize for PCA as
$$\max_{\mathbf{v}} \ \mathbf{v}^T\Sigma\mathbf{v}$$
   Since this is a maximization problem, we can use gradient ascent (not descent) to solve this. Suggest an update equation to find the solution. i.e., find $\mathbf{v}^{k+1}$ from $\mathbf{v}^k$
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   (d) None of these

   (Note: the constraint $||\mathbf{v}|| = 1$ can be enforced as a separate step in each iteration.)

9. Given a set of 2D points X on a line that makes 45 degree to the x-axis:
$$X = \{[-2, 2]^T, [-3, 3]'[-4, 4]^T, [-5, 5]^T[-6, 6]^T\}$$
   We compute the covariance matrix, and its eigen values and eigen vectors. Then identify the correct answer.
   (a) $\lambda_2 = 0$
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10. Consider a loss/objective function that we want to minimize for obtaining w^* as

$$w^* = \arg\min_{w} \ \mathcal{L}(w)$$

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We want to find the minima of the function using gradient descent method. Write update equation for computing w^{k+1} from w^k, where k is the iteration number.

Ans (i): ____________

If $w^0 = 2.0$, what should be w^1 and w^2 if the learning rate $\eta = 0.1$?

Ans (ii): $w^1 =$ ____________ and $w^2 =$ ____________

At the optima, what will be the loss value, i.e., $\mathcal{L}(w*)$

Ans (iii): $\mathcal{L}(w^*) =$ ____________ [1.5 + 3.0 + 3.0 = 7.5]

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- Questions 1-10: 3 Points; Questions 11-12: 7.5 Points. Total 45 Points †

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1. Consider the vector $\mathbf{x} \in R^4$ as $[-8, 6, 0, 0]^T$. Find the following norms:
   (i) L_0 ————
   (ii) L_1 ————
   (iii) L_2 ————
   (iv) L_∞ ————

2. Solution to the optimization problem: "Maximize $\mathbf{w}^T\mathbf{X}\mathbf{X}^T\mathbf{w}$ such that $||\mathbf{w}||_2^2 = 1$" is

   (A) ————————————- Eigen Vector of (B) ————————————

   Fill (A) and (B) with appropriate options from below.

   (A) (i) Lartgest (ii) Second Largest (iii) Smallest (iv) Second Smallest

   (B) (i) $\mathbf{X}^T\mathbf{X}$ (ii) $\mathbf{X}\mathbf{X}^T$ (iii) $\mathbf{X}$ (iv) $\mathbf{X}^T$

3. Consider a dataset of 800 grayscale face images, each of size 120×120 pixels. The faces are frontal and well aligned, and we wish to compute Eigenfaces using PCA. Which of the following statements are true?

   (a) The number of non-zero eigenvalues of the covariance matrix is at most 799.
   (b) A small number (on the order of a few hundred) of eigenfaces can explain most of the variance in the dataset.
   (c) If the dataset consists of images from 10 individuals, with roughly the same number of images per individual, the practical rank of the covariance matrix will be much closer to 10 than to 800.
   (d) Increasing the number of images per individual (while keeping identities fixed) will significantly increase the practical rank of the covariance matrix.
   (e) Applying the same fixed geometric transformation (e.g., horizontal flip or rotation) to all images will not change the rank of the covariance matrix.
   (f) Applying different random geometric transformations independently to each image will tend to increase the practical rank of the covariance matrix.

4. Consider a problem of deciding an email as SPAM or NON-SPAM. We have a trained classifier/solution that scores each email as $\mathbf{w}^T\mathbf{x}$. However, our job will be to define a threshold so that we can say $\mathbf{w}^T\mathbf{x} \geq \theta$ for SPAMs

   For the 10 emails that we have the scores (i.e., $\mathbf{w}^T\mathbf{x}$) are

   $$0.7, 0.65, 0.92, 0.75, 0.45, 0.8, 0.3, 0.8, 0.1, 0.1$$

   and we know the truth/label as (SPAM (+) and NON-SPAM (-))

   $$+, -, +, +, -, +, +, +, -, -$$

   For $\theta = 0.7$, find the "Precision" and "Recall".

   Ans: Precision: ————-; Recall: ————-

5. Consider a dataset ($\mathbf{x}$, y) with Training and Test as:

$$\mathcal{D}_{Train} = \{([+2.0, 0.5]^T, +), ([-3.0, +0.5]^T, -), ([0.0, 3.0]^T, +), ([0.0, -3.0]^T, -)\}$$
$$\mathcal{D}_{Test} = \{([0,0]^T, +), ([+1,+1]^T, +), ([-1,+1]^T, +), ([-1,-1]^T, -), ([+1,-1]^T, -)\}$$

What is the accuracy of a 3-NN classifier with distance function as Euclidean distance? (Euclidean distance between $\mathbf{p}$ and $\mathbf{q}$ is the L2 norm of $\mathbf{p} - \mathbf{q}$

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   Normalize and create a new data set $\mathcal{D}'$ such that mean of this set is $[0,0]^T$

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   (Note: the constraint $||\mathbf{v}|| = 1$ can be enforced as a seperate step in each iteration.)

10. Given a set of 2D points X on a line that makes 45 degree to the x-axis:

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   We compute the covariance matrix, and its eigen values and eigen vectors. Then identify the correct answer.
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$$\mathbf{w}^* = (\mathbf{X^T X})^{-1}\mathbf{X}^T\mathbf{y}$$

In a practical problem, Raju collected data $\mathcal{D} = \{(x, y)\}$ as {(1,3), (2,4), (3,5)}. (note that x is a simple scalar; and on a quick look, you know the solution as $y = x + 2$. Let us verify what we learned in the class really give the same answer.)

By constructing $\mathbf{X}$ and $\mathbf{y}$, compute $\mathbf{w}^*$

Rough Work (will not be graded)